

Applied Thermodynamic II

ME2402

Chapter 01

Heat Transfer

Lecture 02



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Recap

1.2 Conduction Heat transfer

Electrical analogy for composite wall etc..

Heat Flow (Q) > Analogous > Electric Current (I)

Temperature Difference ($t_1 - t_2$) > analogous > Potential Difference (V)

Thermal Resistance > analogous > Electrical Resistance (R)

Electrical representation of heat transfer between solids as follows

So, the Thermal Resistance (R);

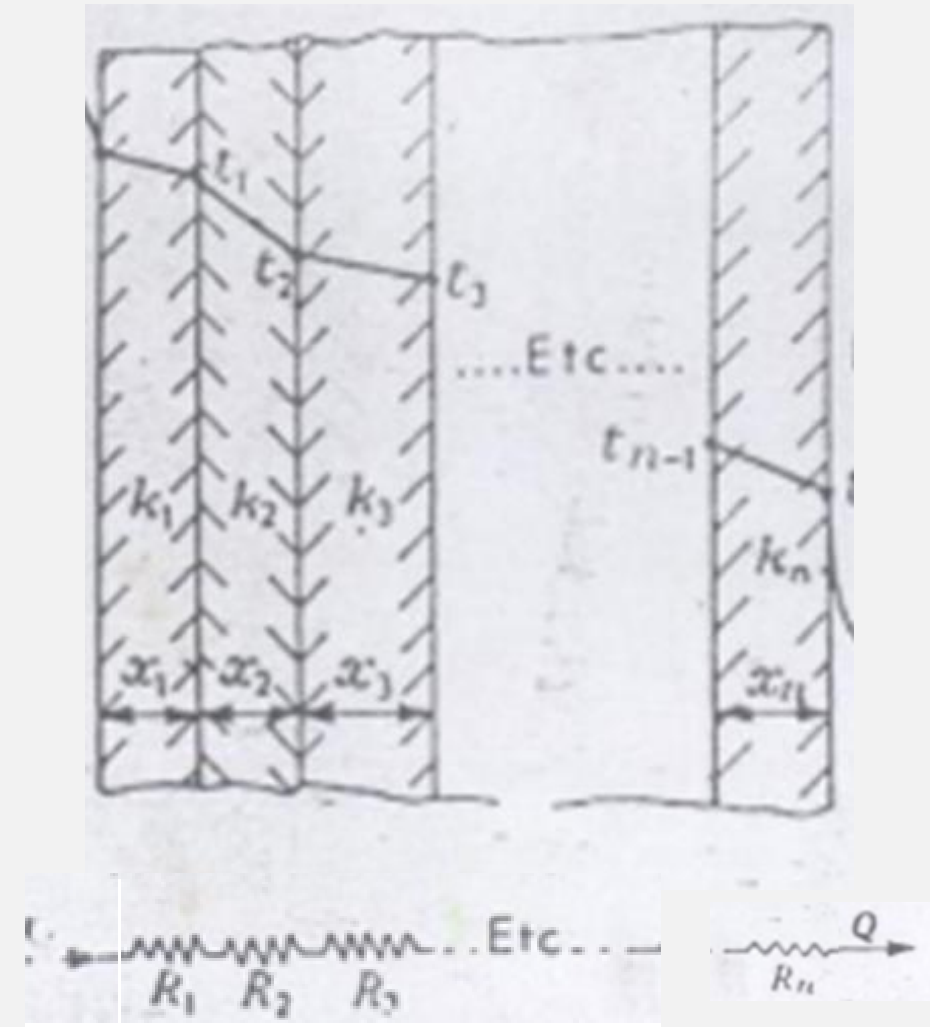
$$R = \frac{x}{kA}$$

For the total composite wall we consider electrical circuit approach

Total resistance (R_T) = $R_1 + R_2 + R_3 \dots \dots + R_n$

Using Furrier equation for the total wall;

$$\dot{Q} = \frac{(t_1 - t_n)}{R_T}$$



1.2 Conduction Heat transfer

Electrical analogy for composite wall etc..

$$\dot{Q} = \frac{(t_1 - t_n)}{R_T}$$

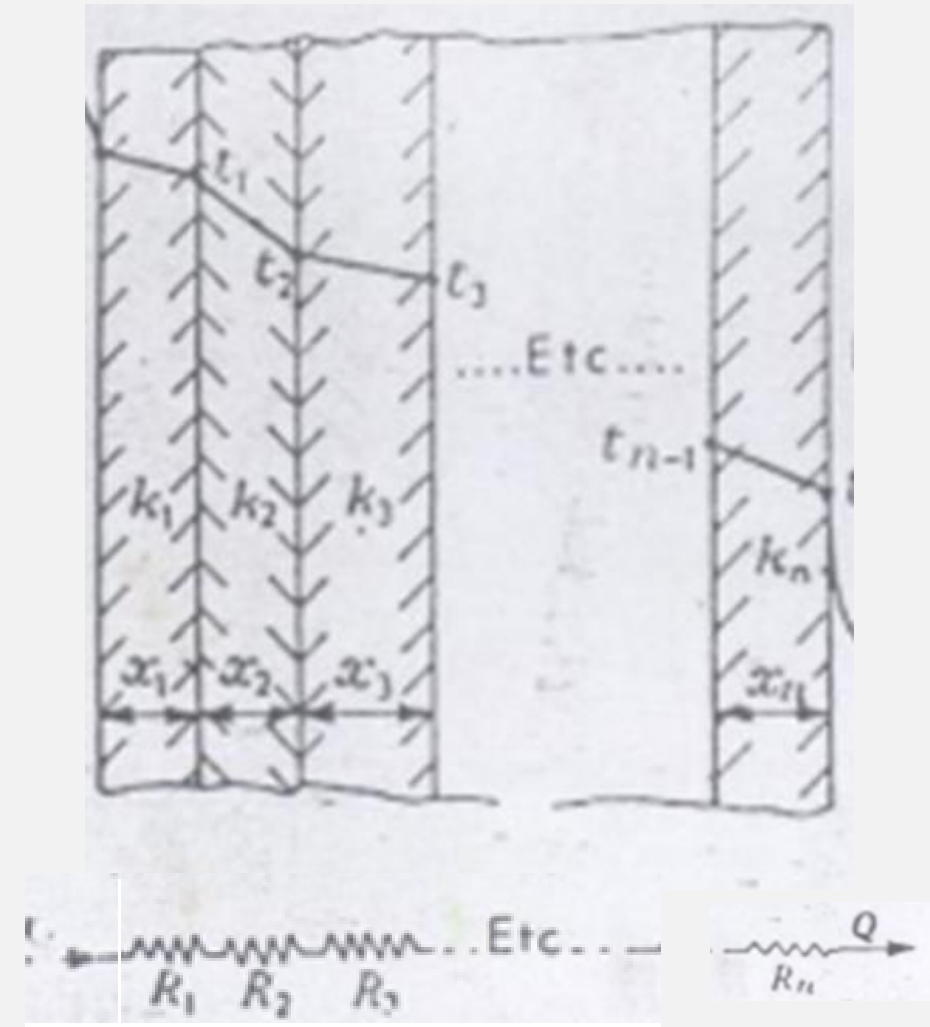
Overall Heat Transfer Coefficient

Define new term called “Overall Heat Transfer Coefficient” given by U

$$\dot{Q} = UA(t_1 - t_n)$$

So, we can derive an relation with R_t and U

$$U = \frac{1}{R_T A}$$

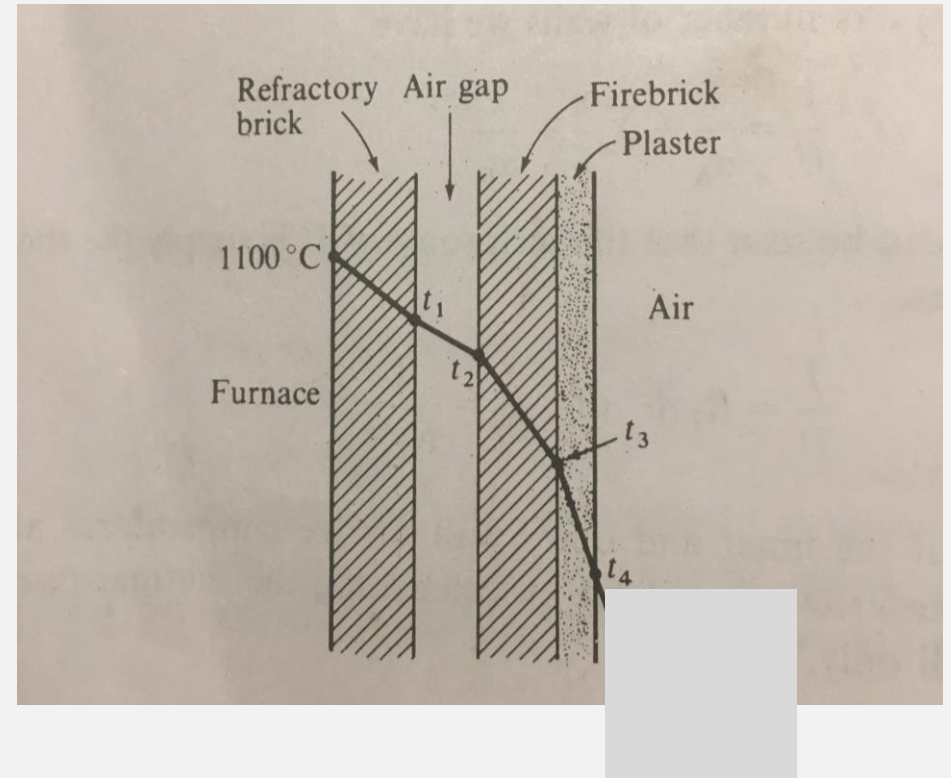


1.2 Conduction Heat transfer

Electrical analogy for composite wall etc..

Example 03

From Previous lecture



1.3 Conduction & Convection Heat transfer from Wall

Newton's Law of Cooling

This approach discuss the heat transfer from one fluid to a solid. This is different from the normal convection.

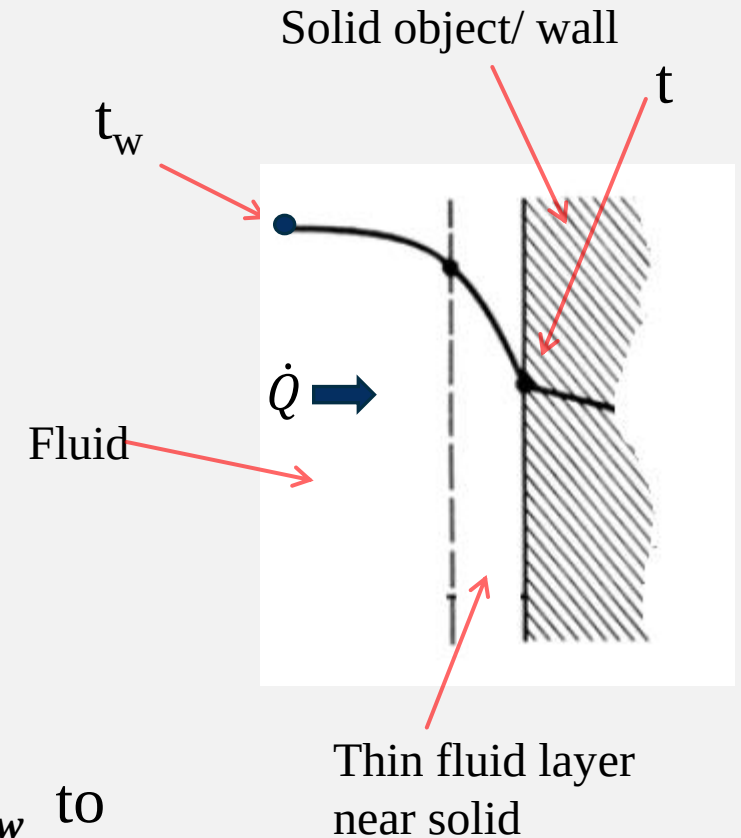
- Around the solid object fluid create a stationary thin film where only conduction happen
- Other region convection, conduction happens

Newton's law discuss the whole region as follows;

Heat transfer from a solid surface with area A at a temperature t_w to a fluid at a temperature t is given by;

$$\dot{Q} = hA(t_w - t)$$

Here h is the Heat Transfer Coefficient between fluid and the solid



1.3 Conduction & Convection Heat transfer from Wall

Electrical analogy to Newton's Cooling

Heat transfer between a solid and a fluid can be represented in electrical form as follows;

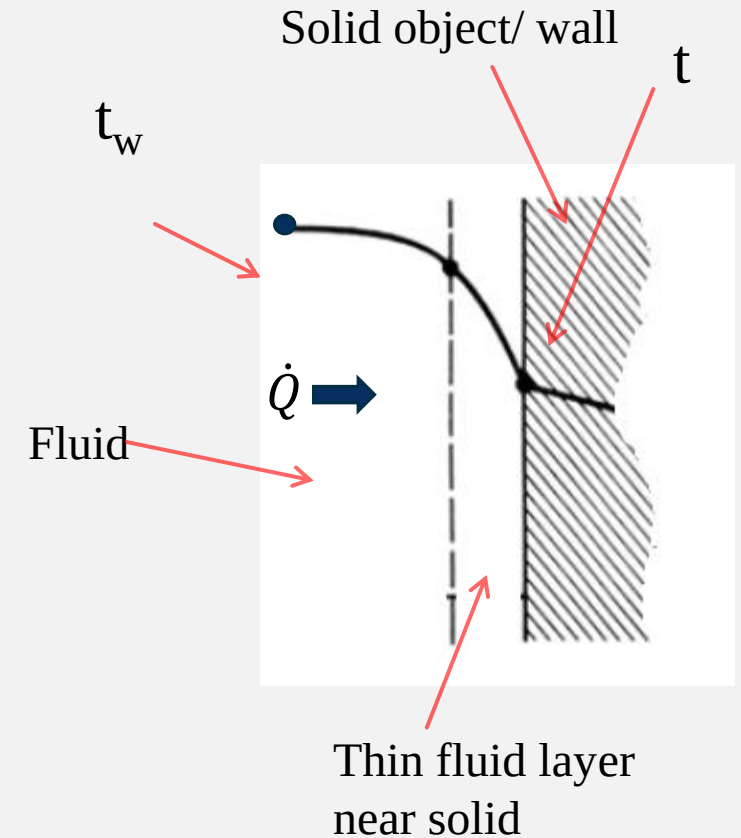
$$\dot{Q} = hA(t_w - t)$$

This is analogous to $i = \frac{v}{R}$ equation in electricity

$$\dot{Q} = \frac{(t_w - t)}{R}$$

So,

$$R = \frac{1}{hA}$$

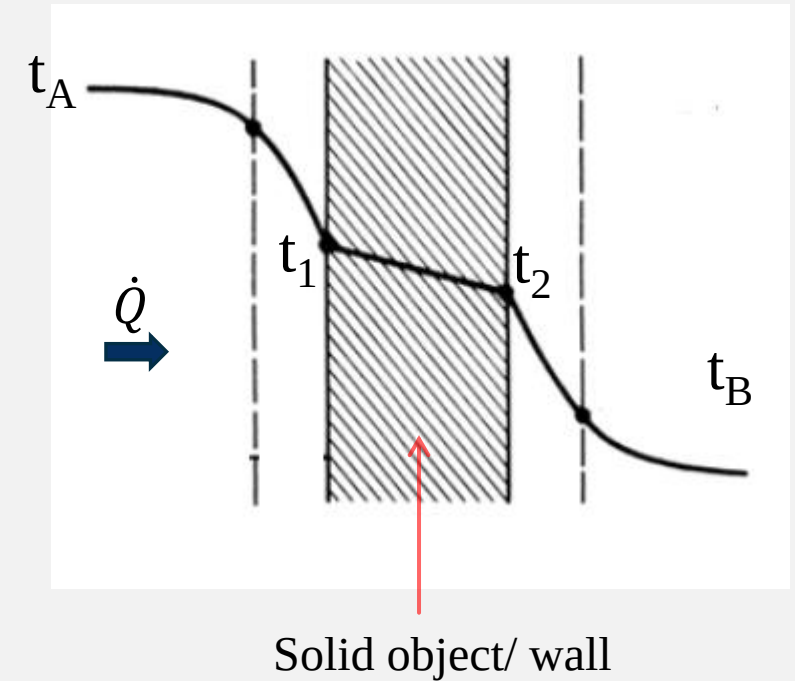


1.3 Conduction & Convection Heat transfer from Wall

Newton's Law of Cooling

Example 01:

A solid wall is having a thermal conductivity of k . Side A is having air at temperature t_A and side B is having air at temperature t_B . Heat transfer coefficients are h_A and h_B respectively. Find an expression for rate of heat transfer

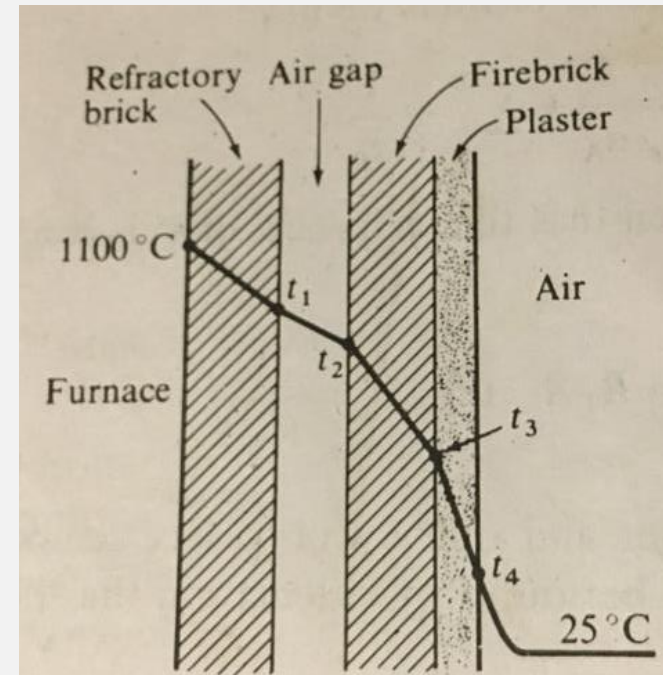


1.3 Conduction & Convection Heat transfer from Wall

Example 02:

Furnace wall consists of 125mm wide refractory brick and 125mm wide insulating firebrick separated by an air gap. The outside wall is covered with a 12mm plaster. The inner surface of the wall is at 1100°C and the room temperature is 25°C . The heat transfer coefficient from the outside wall surface to the air in the room is $17\text{ W/m}^2\text{K}$, and the resistance to heat flow of the air gap is 0.16 K/W . The thermal conductivities of the refractory brick, insulating brick and plaster are 1.6 , 0.3 and 0.14 W/mK respectively. Calculate ;

- The rate of heat loss per unit area of wall surface
- The temperature at each interface
- The temperature at the outside surface of the wall



1.3 Conduction & Convection Heat transfer from Wall

Example 03:

Home work

5. A furnace wall is made of two layers of bricks. Inner refractory brick layer is 130mm and outer insulating brick layer of 130mm. The outside of the furnace wall is covered with an 11mm plaster layer. The temperature of the inner surface of the wall is reaching 1100°C . The furnace is operated in 25°C room temperature condition. The heat transfer coefficient from outside wall to the air in the room is $18\text{W/m}^2\text{K}$. Thermal conductivities of the refractory brick, insulating brick and plaster are 1.6, 0.26, and 0.13 W/m K respectively.
- a) i) Calculate rate of heat loss per unit area of wall surface. (03 marks)
- ii) The temperature at each interface through the wall. (03 marks)
- iii) Temperature of the outer surface of the plaster. (02 marks)
- b) It is suggested to reduce the temperature of the outer surface of the plaster without compromising the performance of the furnace.
- i) Calculate the temperature of the outer surface of the plaster if an air gap with heat resistance of 0.16 K/W is introduced between two brick layers. (04 marks)
- ii) Predict results if the air gap is filled with the same plaster material. (04 marks)
Explain your answer

1.4 Heat transfer through cylinder and sphere

Through Cylindrical Surface

This is widely used in industrial applications

- In heat exchangers
- To insulate thermal fluid lines. Etc....

Consider a pipe with

Internal Radius r_1

External Radius r_2

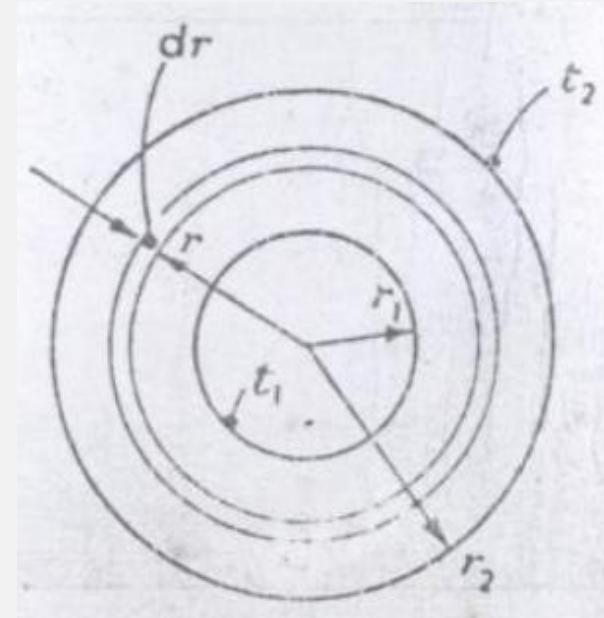
Inside Surface Temperature t_1

Outside Surface Temperature t_2

Thermal Conductivity of Cylinder Material k

Let:

The heat flow through a small element, thickness dr , at a radius of r , where the temperature is t .



1.4 Heat transfer through cylinder and sphere

Through Cylindrical Surface

Let:

The heat flow through a small element, thickness dr , at a radius of r , where the temperature is t .

$$Q = -kA \frac{dt}{dr}$$

$$Q = -k(2\pi r \times 1) \frac{dt}{dr}$$

$$\frac{Q}{r} dr = (-2\pi k) dt$$

$$Q \int_{r_1}^{r_2} \frac{1}{r} dr = -2\pi k \int_{t_1}^{t_2} dt$$

$$Q \log_e \frac{r_2}{r_1} = 2\pi k(t_1 - t_2)$$

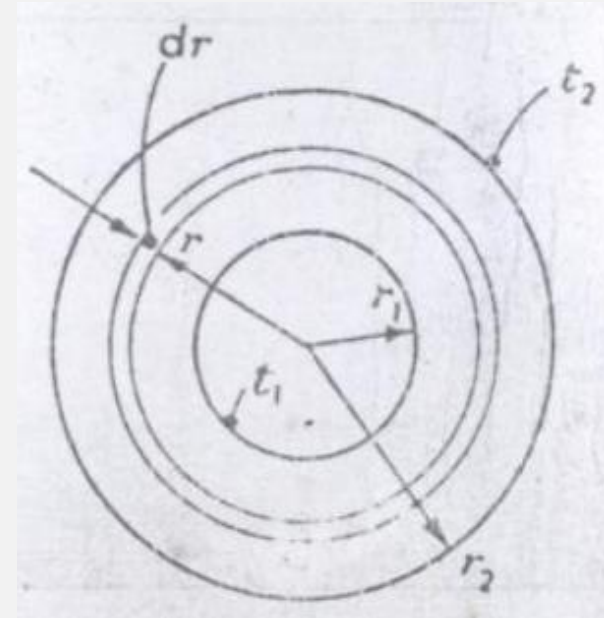
$$Q = \frac{2\pi k(t_1 - t_2)}{\log_e \frac{r_2}{r_1}}$$

Define ;

A_m as $(r_2 - r_1)$;

Then use Furrier equation

$$Q = \frac{kA_m(t_1 - t_2)}{r_2 - r_1}$$



1.4 Heat transfer through cylinder and sphere

Through Cylindrical Surface

$$Q = \frac{2\pi k(t_1 - t_2)}{\log_e \frac{r_2}{r_1}}$$

$$Q = \frac{kA_m(t_1 - t_2)}{r_2 - r_1}$$

Compare above two equations for the same cylindrical surface and get following results

$$A_m = \frac{2\pi (r_2 - r_1)}{\ln \frac{r_2}{r_1}} = 2\pi r_m = \frac{A_2 - A_1}{\ln \frac{r_2}{r_1}}$$

$$r_m = \frac{(r_2 - r_1)}{\ln \frac{r_2}{r_1}}$$

Here ;

A_m is Logarithmic mean area

r_m is Logarithmic mean radius

